

ONLINE ACTIVITY #1

Personalized Linux, Git Commands, and Git Architecture Cheat Sheet

Quick guide: This one-page cheat sheet summarizes essential Linux and Git commands. Each entry includes a short description and a sample command.

1. Basic Linux Commands (10)

Command	What it does	Sample usage
pwd	Shows the full path of the current working directory.	pwd
ls	Lists files and folders in the current directory.	ls -la
cd	Changes the current directory.	cd Documents
mkdir	Creates a new directory.	mkdir project
touch	Creates an empty file or updates a file timestamp.	touch notes.txt
cp	Copies a file or directory.	cp notes.txt backup.txt
mv	Moves or renames a file or directory.	mv old.txt new.txt
rm	Removes a file; use carefully because deletion can be permanent.	rm notes.txt
cat	Displays the contents of a text file.	cat notes.txt
grep	Searches text for lines that match a pattern.	grep "error" log.txt

2. Basic Git Commands (10)

Command	What it does	Sample usage
git init	Creates a new Git repository in the current folder.	git init
git status	Shows changed, staged, and untracked files.	git status
git add	Stages file changes for the next commit.	git add .
git commit	Saves staged changes as a commit with a message.	git commit -m "Add login"
git diff	Shows differences between file versions or changes.	git diff
git restore	Restores working-tree files or removes changes.	git restore app.py
git branch	Lists, creates, or deletes branches.	git branch feature-ui
git checkout / git switch	Moves to another branch; git switch is the modern branch command.	git switch main
git merge	Combines another branch into the current branch.	git merge feature-ui
git tag	Marks a specific commit with a version label.	git tag v1.0
git clone	Copies an existing remote repository to your computer.	git clone <repository-url>

3. Three-Tier Architecture of Git

1. Working Directory – where you edit and create files.
2. Staging Area (Index) – where selected changes are prepared using git add.
3. Local Repository (.git) – where committed project history is stored using git commit.



REFERENCES

- **Official Git Documentation** — Git documentation and command reference.
- **GNU/Linux Command-Line Documentation** — Linux command reference.
- **Course materials and instructor-provided resources.**

AI USE DECLARATION

"The author used ChatGPT to assist with researching and organizing the content of this cheat sheet. The author reviewed and edited the content and takes full responsibility for the final content of the work."